



Style transformation and distance map guided nucleus instance segmentation via multi task learning

Deboch Eyob Abera ^{a,b}, Nazari Zaki ^c, Wenjian Qin ^{a,b},*

^a Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, 1068 Xueyuan Avenue, Shenzhen University Town, Shenzhen, 518055, Guangdong, China

^b University of Chinese Academy of Sciences, No. 19(A) Yuquan Road, Beijing, 100049, Beijing, China

^c Department of Computer Science and Software Engineering, College of Information Technology, United Arab Emirates University, Al Ain, Abu Dhabi, United Arab Emirates

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ABSTRACT

Instance segmentation of nuclei in histopathological images is hindered by three critical challenges: overlapping nuclei, domain shift caused by staining variability, and generalization across diverse multi-organ datasets. To address these issues, we propose a unified multi-task learning framework for nucleus instance segmentation that integrates style transformation and distance map-guided segmentation. Our architecture employs multi-dilated residual blocks and encoder-decoder attention gates to capture multi-scale features and preserve fine nuclear details, while a transformer in the bottleneck enhances contextual understanding and models long-range dependencies. The network incorporates dual heads for semantic segmentation and distance-map prediction, effectively addressing overlapping nuclei. Additionally, a histogram-based, reference-guided stain normalization module mitigates domain shift caused by staining variability, and when combined with our robust model architecture, it enhances the overall generalization ability across multi-organ datasets. Experimental results demonstrate our method's superior performance over existing segmentation approaches. The source code is available at <https://github.com/eyob12/MTL-NucleusSeg>.

1. Introduction

In computer vision, image segmentation refers to highlighting targeted regions of interest from an image. In histopathology, particularly in H&E-stained images, the segmentation of nuclei is crucial for disease identification and diagnosis [1,2]. Histopathologists typically focus on nuclei to extract key information for diagnosis, but manual identifying and analyzing nuclei can be time consuming and repetitive [3,4]. This is why automated segmentation techniques are being developed to make the process faster and more reliable [5]. There are usually two kinds of segmentations that histopathologists are interested in for H&E-stained images. The first is semantic segmentation, where each pixel in an image will be labeled as either nuclei or background. This segmentation helps histopathologists in H&E-stained images analysis by simplifying the focus on nuclei and removing potential distractions from the background. However, semantic segmentation, can only give histopathologists a very limited information [6]. The second is instance segmentation where every instance of the foreground is uniquely labeled, which means every single nucleus in the case of nuclei segmentation must have its own label. This task is much more complicated

and challenging than semantic segmentation because some nuclei are positioned so closely together that it becomes difficult to detect them as separate instances [7,8]. However, instance segmentation provides significantly more detailed information than semantic segmentation since it allows for the identification of the number of nuclei present in a given digital H and E slide [9,10]. In Fig. 1, detailed information is provided regarding the current challenge in histopathological imaging including a brief introduction to semantic segmentation—where each pixel is labeled as nucleus or background; instance segmentation—where overlapping nuclei are uniquely separated and color-coded; the overlapping nuclei problem—which semantic segmentation cannot resolve but instance segmentation solves by uniquely separating nuclei; domain shift—shown by variations in image appearance between organs causing the model's accuracy to drop; and generalization across different organs—shown by a single model handling multiple tissue types and producing consistent instance-segmentation results.

Segmentation of nuclei in histopathological images is essential but highly challenging due to several inherent limitations. Nuclei in tissue

* Corresponding author at: Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, 1068 Xueyuan Avenue, Shenzhen University Town, Shenzhen, 518055, Guangdong, China.

E-mail addresses: deboch@siat.ac.cn (D.E. Abera), nzaki@uaeu.ac.ae (N. Zaki), wj.qin@siat.ac.cn (W. Qin).

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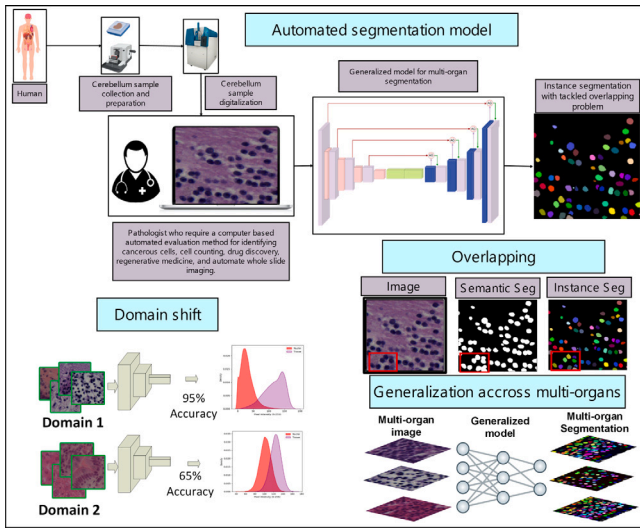


Fig. 1. Overview of sample collection and preparation, semantic segmentation, instance segmentation, and key challenges in histopathological imaging, including overlapping nuclei segmentation, domain shift and generalization across diverse multi-organ datasets.

samples are often densely packed, leading to issues such as overlapping and merging. Variability in imaging conditions, resolutions, and tissue types further complicates the problem, as models trained on one dataset often fail to generalize well to another datasets. Moreover, differences in staining intensity across slides introduce additional variability, negatively impacting model performance. These challenges demand a robust and innovative instance segmentation approach capable of effectively addressing these issues [10,11]. Despite the progress made by existing studies, nuclei segmentation is hampered by three significant obstacles, such as the overlapped problem, domain shift, and the challenge of generalizing across diverse nuclei images. Fig. 1 provides a brief overview of these challenges in histopathological imaging.

Rethinking current challenges, we propose a unified multi-task learning framework that integrates style transformation-based stain normalization with distance map-guided segmentation. This approach resolves overlapping nuclei, mitigates domain shift, and generalizes across diverse organs. Specifically, to tackle overlapping nuclei, our model employs a dual-head design that jointly predicts semantic masks and distance maps. The distance map head guides the precise separation of densely clustered nuclei. Our architecture introduces multi-dilated residual blocks with varying dilation rates and encoder-decoder attention gates to effectively capture multi-scale features and optimize feature extraction through skip connections, thereby preserving fine-grained nuclear details. Furthermore, we incorporate a transformer in the bottleneck layer to improve contextual awareness and preserve long-range dependencies for complex nuclear structures. To address domain shift, and the challenges of generalizing across diverse histopathological datasets, we propose an innovative approach that leverages stain normalization techniques through style transfer. This method transforms the original images to align with reference samples that exhibit staining variations commonly encountered in real-world medical imaging. By incorporating this style transfer into the training process along with a set of robust augmentation techniques, our model learns to handle the diversity and variability of histopathological data. This approach significantly improves the model's generalization capability, allowing it to perform more effectively across multiple datasets with different staining protocols and imaging conditions. Overall, our proposed approach enables precise and automated nuclei segmentation across diverse histopathological images, enhancing both its generalization capabilities and its ability to handle overlapping and

instance segmentation challenges. The contributions of this work are summarized as follows:

1. We introduce a unified multi-task learning framework that combines semantic segmentation and distance map prediction, enabling accurate separation of densely packed and overlapping nuclei in histopathological images.
2. We design an advanced network architecture that integrates multi-dilated residual U-Net blocks, attention gates in every skip connection, and a bottleneck transformer, which together enhance multi-scale feature extraction, improve contextual understanding, and enable more precise instance segmentation.
3. We propose a novel reference-guided stain normalization strategy, using histogram-based style transformation and robust augmentation, which effectively addresses staining variability and domain shift, thereby improving generalization across diverse multi-organ datasets.
4. Demonstrated state-of-the-art performance and generalization across multiple challenging benchmarks (NuInsSeg, MoNuSeg, DSB2018), outperforming existing baseline and SOTA methods in both semantic and instance segmentation metrics.

2. Related work

Nucleus segmentation techniques in histopathology have evolved significantly over time. Traditional image processing-based approaches have historically been used for nucleus segmentation, relying on manual feature extraction and predefined rules to process images [12–14]. Despite their widespread use, traditional image processing methods have several limitations when applied to complex and heterogeneous biological datasets. One of the primary limitations of traditional methods is their dependency on hand-crafted features. These methods often require careful tuning of parameters and are heavily reliant on the explicit design of features that can robustly distinguish between objects and background. Biological images, particularly those in histopathology, exhibit high variability due to factors such as tissue heterogeneity, uneven staining, and overlapping structures. As a result, traditional segmentation methods can struggle with accurately delineating objects, particularly when there are variations in object size, shape, and intensity. To address these limitations, machine learning methods, particularly convolutional neural networks (CNNs), have been widely adopted for medical image segmentation. In the field of nuclei instance segmentation, CNN-based approaches can be classified into three categories: detection-based models like Mask R-CNN [15], encoder-decoder frameworks like U-Net and its derivatives, and multi-tasking neural networks.

The detection-based models rely on a Region Proposal Network (RPN) to identify regions in an image that are likely to contain objects of interest, or foreground objects. One of the most notable detection-based approaches is Mask R-CNN [16], which is built on top of Faster R-CNN, a model that itself improved upon the earlier R-CNN and Fast R-CNN architectures [17,18]. Mask R-CNN consists of three main components. The first is the RPN, which proposes candidate regions in the image likely to contain objects. These regions are then processed by a two-branch convolutional neural network. The first branch focuses on generating bounding boxes for all detected instances of foreground objects, while the second branch, a fully convolutional network, produces an instance segmentation binary mask for each identified object within the proposed region. The Segment Anything Model (SAM) can also be categorized as a detection-based approach [19]. Unlike traditional models, SAM employs prompts (such as points, bounding boxes, or textual descriptions) to guide its decoder in isolating and segmenting the foreground object of interest. One of the limitations of these approaches is their heavy dependence on region proposals, which makes it challenging to accurately segment closely packed nuclei.

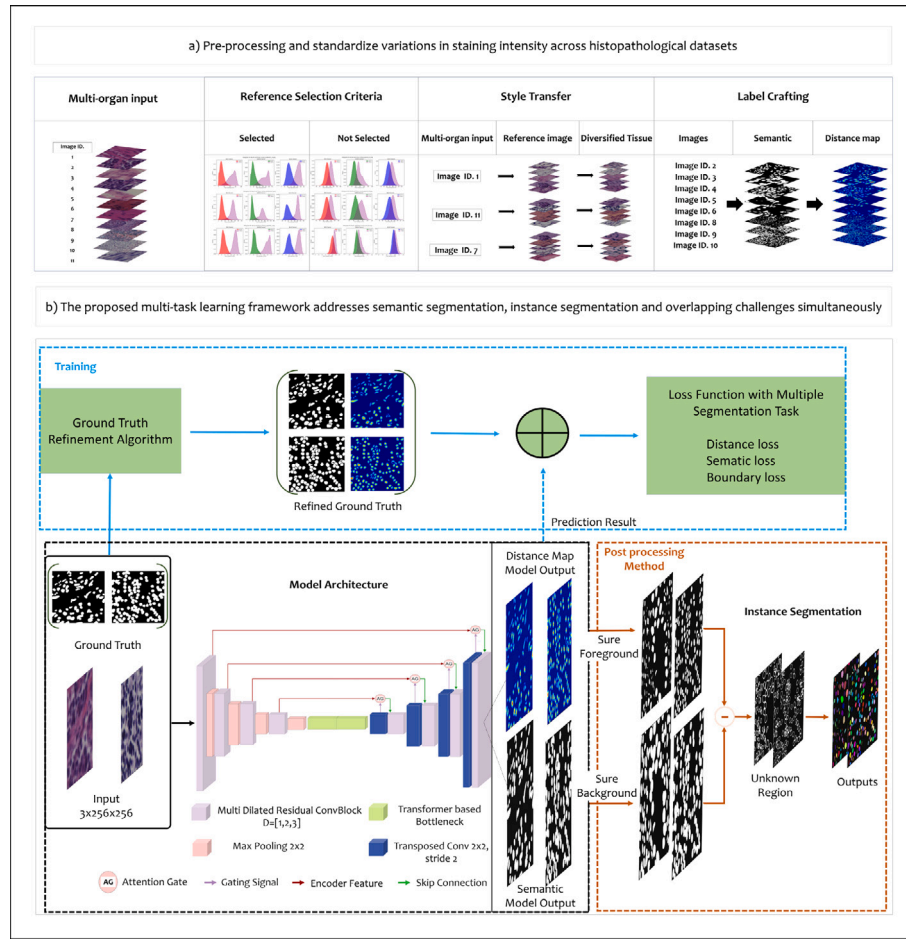


Fig. 2. Overview of the proposed methodology for multi-organ nucleus segmentation, including multi-organ data preparation, stain normalization using selected reference style transfer and label crafting, multi-task model training, and instance segmentation via post-processing.

In encoder–decoder frameworks [20,21], Convolutional Neural Networks (CNNs) are the backbone of the model. U-Net stands out for its encoder–decoder structure with skip connections, which effectively recover spatial information lost during downsampling. Modified versions of U-Net, such as U-Net++ [22], and R2U-Net [23], further enhance segmentation accuracy through improved skip connections and residual learning. However, Most of the CNN architectures, such as SegNet [24] or U-Net [21], are not sufficient on their own for achieving instance segmentation. These architectures typically require additional post-processing using traditional image processing techniques. These post-processing methods depend heavily on specific types of outputs generated by the CNNs. As a result, various approaches have been developed to craft appropriate labels that can be effectively utilized in the desired post-processing pipeline to achieve instance segmentation. One of the basic techniques is using morphological operations, such as dilation and erosion, to refine segmentation masks, remove noise, and separate connected components [25,26]. However, these techniques are not effective for complicated cases like densely packed histopathological nucleus images, as they attempt to generate instance segmentation from a single semantic output through postprocessing.

In multi-tasking neural network frameworks, multiple outputs are processed simultaneously for effective solution of instance nuclei segmentation, but the success of reliable instance segmentation relies on the collaborative integration of model design and label crafting [7, 27,28]. Over the years, various hand-crafted label techniques have been developed to support instance segmentation tasks. These include marker center [29], distance maps [8], StarDist approaches [27], HoVerNet approaches [7], gradient tracking [30], graph-based methods [31], contour detection [32], and interval maps [28]. These labels

are specifically designed to enhance the segmentation process, often integrating seamlessly with post-processing methods such as the watershed algorithm or other advanced techniques. However, the performance of multi-task approaches is highly dependent upon robust architectural designs, optimal integration of label representations, and careful balance of multiple task losses during training [9,33]. Moreover, despite their sophistication, these methods still face generalization challenges across diverse datasets due to staining variability and domain shifts. Our proposed framework addresses these limitations by combining multi-task learning with reference-guided stain normalization. Unlike existing methods, our approach explicitly tackles the issues of densely overlapping nuclei and staining-induced domain variability, enhancing both segmentation accuracy and cross-dataset generalization capabilities.

3. Proposed method

The proposed methodology for multi-organ nucleus segmentation addresses the inherent challenges of histopathological image analysis, particularly dense packing, overlapping structures, generalability, and variability in staining across diverse datasets. The methodology, as illustrated in Fig. 2, is structured into two primary stages: (1). Pre-Processing for Standardization Variability: This stage aims to normalize variations in staining intensity and enhance the quality of histopathological images, ensuring consistency across datasets. Key components include reference selection criteria to identify representative reference images, style transfer techniques to harmonize visual characteristics, and label crafting to prepare reliable training data.

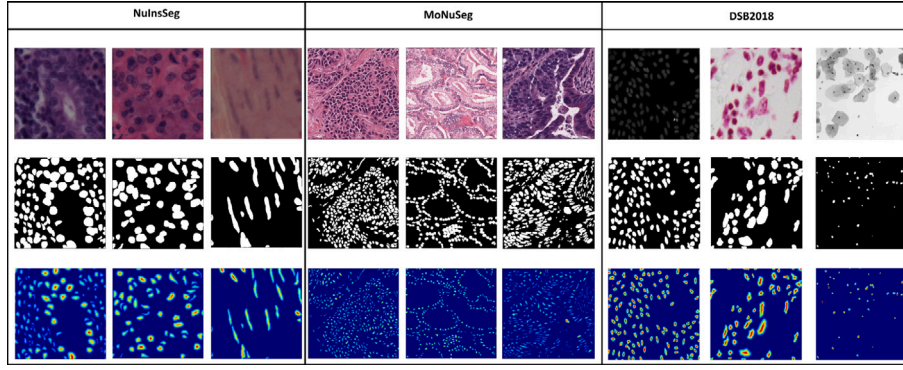


Fig. 3. Representative examples from NuInsSeg, MoNuSeg, and DSB2018 datasets: original images (first row), ground truth masks (second row), and refined and generated distance weight maps (third row).

These pre-processing steps mitigate variability caused by diverse imaging conditions, enabling the model to generalize effectively across heterogeneous datasets. (2). **Proposed Model Architecture:** The model employs a multi-task learning framework to simultaneously generate two outputs that facilitate accurate nucleus segmentation. First, a distance map $D \in \mathbb{R}^{H \times W}$ encodes spatial relationships, which guide in the separation of overlapping nuclei. Second, a semantic segmentation map $S \in \mathbb{R}^{H \times W}$ classifies each pixel as either nucleus or background. This dual-output approach leverages the complementary nature of the two tasks, improving the delineation of individual nuclei while maintaining high segmentation accuracy. These two key stages of the methodology are detailed in the following five sections.

3.1. Dataset pre-processing and ground truth refinement

In this research, we used three publicly available multi-organ nuclei segmentation datasets: NuInsSeg [34], MoNuSeg [35], and DSB2018 [36]. Fig. 3 shows representative examples from the three datasets. To enhance model training, we performed additional distance map ground truth by generating Euclidean distance maps derived from the provided manual annotations. The **NuInsSeg** dataset is a large-scale, manually annotated benchmark designed for nucleus instance segmentation. It consists of 665 image patches collected from 21 human organs and 10 mouse organs, resulting in a total of 30,698 annotated nuclei. All images are of equal size and were captured at 40 $\times$ magnification. This dataset does not include classification labels for different nucleus types. For preprocessing, all images were resized to $3 \times 256 \times 256$ pixels to standardize input dimensions for our model. The **MoNuSeg** dataset includes 37 training and 14 test images, each with a resolution of $3 \times 1000 \times 1000$ pixels. The images are sourced from seven organs and were acquired at 40 $\times$ magnification. The training set contains approximately 21,623 annotated nuclei, while the test set contains around 7223. To ensure consistent input size, each image was divided into 16 non-overlapping patches of size $3 \times 256 \times 256$ pixels. The **DSB2018** dataset was introduced in the Data Science Bowl competition as a comprehensive benchmark for nucleus segmentation. It includes 670 microscopy images with about 30,000 manually annotated nuclei. The dataset comprises images from a wide range of biological experiments and includes both brightfield and fluorescence modalities. It features various staining techniques such as DAPI and H&E, and includes samples from different species including human, mouse, and fly. This variability makes it suitable for training robust models that can generalize across tissue types and imaging conditions.

In summary, we performed standardized preprocessing across all datasets. This included resizing or patching all images to $3 \times 256 \times 256$ pixels and generating distance map ground truth labels to support instance-level learning during training.

3.2. Reference-based style transfer with augmentation for domain adaptation

After pre-processing the dataset, we propose a reference-based style transfer and augmentation to standardize variations in staining intensity across histopathological datasets. Fig. 4 illustrates the flowchart of our proposed approach. Reference images for stain normalization are selected based on histogram analysis. Images where the bell-shaped histograms of the foreground (nuclei) and background (tissue) overlap significantly are excluded, as they lack clear stain distinction. Conversely, images with distinct foreground and background histograms are chosen as references to ensure better separation between nuclei and tissue. The style of all images in the dataset is then transferred from these selected reference images, followed by augmentation techniques such as rotation and flipping to expand the dataset. This preprocessing step increases the dataset size by generating five variations of each image with different stain styles and augmentations, helping the model to effectively address the domain generalization problem. To mathematically represent the reference-based style transfer algorithm, we assume that I be an input histopathological image, and R be a reference image selected based on histogram analysis criteria. The first step is convert both images I and R from RGB to LAB color space: Then, for each channel $k \in \{L, A, B\}$, compute the mean μ and standard deviation σ :

$$\mu_I^{(k)} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W I(i, j, k), \quad (1)$$

$$\sigma_I^{(k)} = \sqrt{\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (I(i, j, k) - \mu_I^{(k)})^2}$$

$$\mu_R^{(k)} = \frac{1}{H' \times W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} R(i, j, k), \quad (2)$$

$$\sigma_R^{(k)} = \sqrt{\frac{1}{H' \times W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} (R(i, j, k) - \mu_R^{(k)})^2}$$

Next, for each pixel (i, j) in the input image and channel k , apply the affine transformation:

$$N(i, j, k) = \text{clip} \left(\frac{\sigma_R^{(k)}}{\sigma_I^{(k)}} \cdot (I(i, j, k) - \mu_I^{(k)}) + \mu_R^{(k)} \right) \quad (3)$$

where $\text{clip}(x) = \max(0, \min(255, x))$ ensures pixel values remain in $[0, 255]$.

Then, the normalized image $N \in \mathbb{R}^{H \times W \times 3}$ can be expressed as:

$$N = \text{clip} \left(\underbrace{\left(\frac{\sigma_R}{\sigma_I} \odot (I - \mu_I) \right)}_{\text{Standardization}} + \mu_R \right), \quad (4)$$

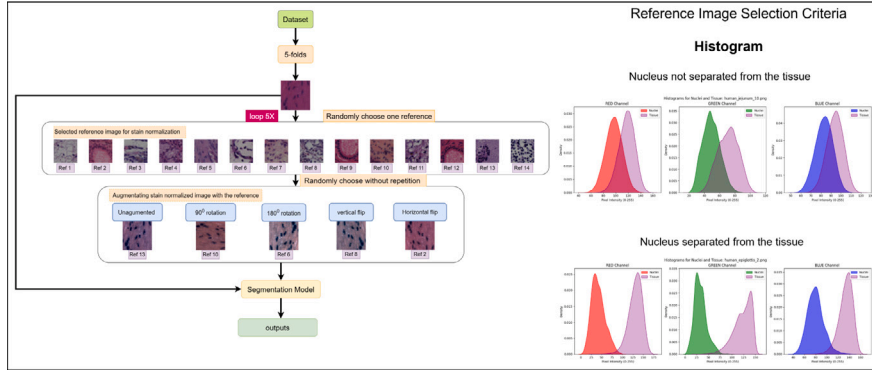


Fig. 4. Flowchart of the proposed training strategy, including dataset preparation, style transfer with reference-based stain normalization, augmentation techniques, and segmentation model training.

where: $\odot$: Element-wise multiplication (broadcasted across channels), μ_I, μ_R : Mean vectors for input and reference, σ_I, σ_R : Std vectors for input and reference.

After normalization, spatial augmentations $\mathcal{A}$ (rotations, flips) are applied stochastically:

$$N_{\text{augmented}} = \mathcal{A}(N), \quad \mathcal{A} \sim \{\text{Rotation, Flip}\}. \quad (5)$$

Finally, The preprocessed image $\hat{I}$ is:

$$\hat{I} = \text{RGB}(N_{\text{augmented}}), \quad (6)$$

where $\text{RGB}(\cdot)$ converts the normalized LAB image back to RGB space. This formulation ensures that the stain distribution of the input image aligns with the reference while preserving spatial features through augmentation.

3.3. Model architecture design

In this study, we propose an advanced multi-task segmentation model: Attention Gate Guided Multi-Dilated Residual Network with Bottleneck Transformer, designed to address the challenges posed by densely packed nucleus instance segmentation (See Fig. 5). The foundation of our architecture is the U-Net framework. The U-Net model is well-regarded for its ability to perform semantic segmentation tasks efficiently by using encoder-decoder structures and skip connections to retain spatial details. To enhance its capabilities, we proposed multi-dilated residual blocks at each layer of the network. The multi-dilated residual blocks are specifically designed to extract multi-scale features, a crucial requirement for segmenting overlapping and densely packed nuclei. The varying dilation rates within these blocks expand the receptive fields of convolutional operations, enabling the model to simultaneously capture fine details and broader contextual information. Furthermore, the residual connections within these blocks address the vanishing gradient problem by allowing gradients to flow through the network more effectively, ensuring robust feature propagation. To refine the feature maps and improve the model's ability to focus on critical regions, we propose attention gates into every skip connection of the Multi-Dilated Residual U-Net framework. Skip connections are crucial in U-Net-based architectures as they bridge the encoder and decoder pathways, preserving spatial information. By embedding attention mechanisms within these connections, the network is guided to prioritize features from relevant regions, such as nuclei, while suppressing background noise. This selective attention ensures that only the most significant features contribute to the final segmentation output, enhancing the overall accuracy and efficiency of the model. To capture global contextual information, we proposed a transformer block in the bottleneck layer of the U-Net. While convolutional operations excel at capturing local patterns, they are inherently limited in their ability to model long-range dependencies within the image. The

transformer block addresses this limitation by leveraging self-attention mechanisms, which allow the network to model relationships between distant regions in the input image. This global perspective is particularly important for distinguishing between closely packed nuclei, as it helps the model understand their spatial relationships and boundaries more effectively. Additionally, the transformer block enriches the feature representations by integrating both local and global information, leading to more accurate segmentation. Attention Gate Guided Multi-Dilated Residual Network with Bottleneck Transformer makes the proposed architecture highly effective for the task of nucleus instance segmentation. The multi-dilated residual blocks enhance feature extraction at multiple scales, the attention gates improve the focus on relevant regions, and the transformer block introduces a global context to the feature representation. Together, these components create a synergistic effect, enabling the model to handle the complexities of densely packed and overlapping nuclei with higher precision. Through ablation studies, we demonstrate the significant performance improvements achieved. Each addition contributes uniquely to the overall performance, validating the importance of our architectural design choices (See Table 2).

3.4. Multi-task learning for segmentation

In our proposed multi-task learning architecture, the model is designed with two output heads to address both semantic segmentation and instance segmentation tasks simultaneously. These outputs are processed in parallel to achieve optimal segmentation performance. The semantic segmentation head is responsible for generating the binary segmentation mask, which distinguishes the foreground (segmented objects) from the background. The distance map head produces the distance map, which is crucial for instance segmentation as it indicates the Euclidean distance from each pixel to the nearest boundary of the segmented object. This helps in distinguishing separate objects, even if they are densely packed.

The total loss for the multi-task model is a combination of three loss components: Binary Cross-Entropy (*BCE*) Loss, Mean Squared Error (*MSE*) Loss, and Gradient Difference Loss (*GDL*). The Binary Cross-Entropy (*BCE*) Loss is used for the binary segmentation mask produced by the semantic segmentation head. This loss focuses on pixel-wise accuracy, penalizing the difference between the predicted binary mask and the ground truth binary mask. Mathematically, *BCE* is defined as:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N (\text{target}_i \times \log(\text{pred}_i) + (1 - \text{target}_i) \times \log(1 - \text{pred}_i)) \quad (7)$$

where pred represents the predicted binary segmentation mask (after sigmoid activation) for the 2D image, target is the ground truth binary mask for semantic segmentation, and N is the number of pixels in the 2D image.

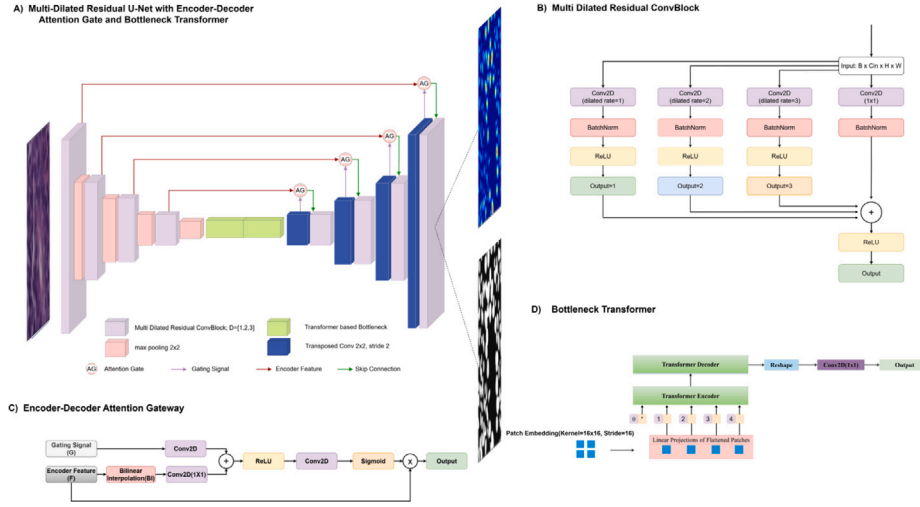


Fig. 5. Proposed attention-based multi-dilated residual U-Net with Transformer for multi-organ nucleus segmentation. The architecture includes multi-dilated residual blocks with dilation rates [1, 2, 3] at each layer, attention gates to enhance focus on relevant regions, and a transformer block in the bottleneck layer to capture long-range dependencies.

The MSE loss function ensures that the predicted distance map preserves the global structure of the objects, maintaining the overall alignment of the pixel intensity with the target distance map. The MSE loss is formulated as:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{HW} \sum_{x=1}^H \sum_{y=1}^W (D_{\text{pred}}(x, y) - D_{\text{target}}(x, y))^2 \quad (8)$$

where H and W are the spatial dimensions of the distance map, and D_{pred} and D_{target} represent the predicted and target distance maps, respectively.

The Gradient Difference Loss (GDL) function enhances the sharpness of object boundaries by penalizing discrepancies in gradient magnitudes between the predicted and target maps. The GDL is computed as:

$$\mathcal{L}_{\text{GDL}} = \frac{1}{HW} \sum_{x,y} (|\nabla_x D_{\text{pred}} - \nabla_x D_{\text{target}}| + |\nabla_y D_{\text{pred}} - \nabla_y D_{\text{target}}|) \quad (9)$$

where ∇_x and ∇_y represent the horizontal and vertical gradients, computed via finite differences. The total loss is a weighted sum of BCE, MSE and GDL:

$$\mathcal{L}_{\text{Total}} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{MSE}} + \lambda_3 \mathcal{L}_{\text{GDL}} \quad (10)$$

where λ_1 , λ_2 and λ_3 are hyperparameters that control the balance between global structure and boundary sharpness. $\lambda_1 = 0.5$, $\lambda_2 = 0.6$ and $\lambda_3 = 0.7$.

3.5. Instance segmentation via postprocessing

To achieve instance segmentation using the outputs from our proposed multi-task model, we employed a distance map-guided watershed post-processing technique, which effectively refines segmentation results, especially for closely packed nuclei. This approach is designed to distinguish individual objects and address the challenges posed by overlapping or touching nuclei. The process, as visualized in Fig. 6, begins with refining the semantic segmentation map and the distance weight map, both of which are outputs of the U-Net model. Next, we generate the “sure background” by identifying regions confidently outside the nuclei from the refined semantic segmentation map. These regions act as the background markers for the watershed algorithm. Simultaneously, we derive the *sure foreground* from the refined distance weight map by extracting regions confidently inside the nuclei, which serve as foreground markers. By comparing the sure background and the sure foreground, we compute an intermediate mask representing

the difference between the two. This intermediate mask is critical for delineating overlapping nuclei and preparing the input for the watershed algorithm. The final step involves applying the watershed algorithm using the intermediate mask as markers and the refined segmentation map as the landscape. This method allows for the accurate separation of individual nuclei, even in densely packed regions, resulting in an instance segmentation map where each nucleus is distinctly segmented. This approach ensures improved segmentation accuracy and addresses the limitations of traditional methods in handling closely packed objects.

4. Experimental results and analysis

4.1. Implementation and training details

We trained our proposed model on the NuInsSeg [34], MoNuSeg [35], DSB2018 [36] datasets using an NVIDIA A6000 GPU. To enhance data diversity, style transfer was applied, followed by augmentation, as described in Section 3.2, increasing the dataset size five times. The reference image selection criteria for style transfer are also detailed in Section 3.2. After data pre-processing and style transformation, the segmentation model was trained with 5-fold cross-validation.

4.2. Evaluation metrics

To evaluate the performance of segmentation models, we employ four evaluation metrics: Dice Score [37], Intersection over Union (IoU) [38], Aggregated Jaccard Index (AJI) [39], and Panoptic Quality (PQ) [38]. These metrics provide comprehensive insights into both semantic and instance segmentation tasks. The Dice score is a measure of overlap that is particularly suited for semantic segmentation. It is sensitive to boundary alignment and evaluates how well the predicted and ground truth masks match [37]. IoU is another metric used in semantic segmentation, measuring the degree of overlap between the predicted and ground truth masks [38]. The AJI metric is tailored for instance segmentation and evaluates the match between predicted instances and ground truth instances [39]. PQ is a comprehensive metric used for instance segmentation, assessing both recognition quality (RQ) and segmentation quality (SQ) [38].

4.3. Qualitative experimental result on multiple dataset

Our proposed model was trained and evaluated on three benchmark datasets: NuInsSeg [34], MoNuSeg [35], and DSB2018 [36]. Figs. 7, 8 and 9 present representative segmentation results from the NuInsSeg,

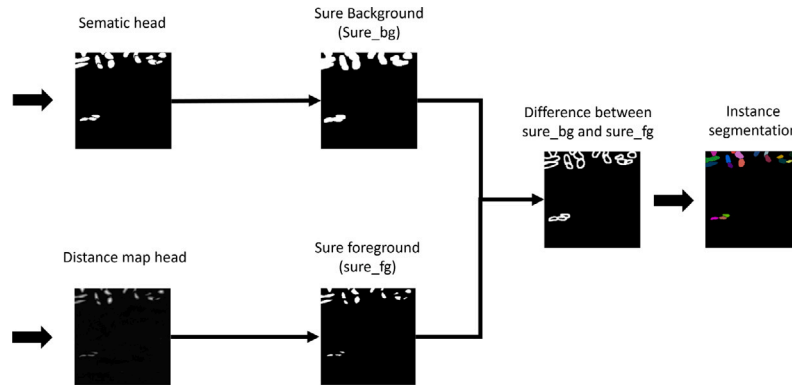


Fig. 6. Overview of the proposed watershed-based post-processing technique for refining and separating closely packed nuclei in instance segmentation.

MoNuSeg and DSB2018 datasets, respectively. The segmentation results in Figs. 7, 8, and 9 confirm the model's ability to accurately segment nuclei across diverse datasets, including those with densely clustered nuclei, without any observable performance loss. These findings validate the efficacy of the proposed approach in achieving superior semantic and instance segmentation performance across multiple datasets. In Fig. 10, we further analyzed the model's performance in overlapping scenarios by examining samples from NuInsSeg, MoNuSeg and DSB2018 across different organs. The results presented in Fig. 10, highlighted by blue boxes, showcase how our model performs in scenarios involving overlapping nuclei. Fig. 10, the first three image samples are from the NuInsSeg dataset, while the remaining come from the MoNuSeg and DSB2018 datasets. The results are organized across multiple columns to clearly illustrate the process and effectiveness of our model. The first column displays the original images, and the blue box highlights the regions where the nuclei are overlapping in H&E-stained slides. The second column shows the ground truth mask boundaries, which are aligned with the original image, demonstrating how the ground truth captures the overlapping nuclei. In the third column, the model's semantic segmentation predictions are shown, where, in some cases, the model struggles to separate the overlapping nuclei, as indicated by the blue box. The fourth column presents the markers derived from the distance map, where the blue box highlights how the markers successfully identify and separate each nucleus individually, even in areas with overlaps. Finally, the last column shows the resolved overlapping scenarios, where the blue box highlights the effective separation of the overlapping nuclei by the model. In conclusion, Fig. 10 demonstrates that the proposed model effectively addresses the challenge of overlapping nuclei while maintaining excellent performance. This result underscores the robustness and adaptability of the model in handling complex segmentation tasks.

4.4. Quantitative experimental result on multiple dataset

This section presents the quantitative evaluation of the proposed model using four commonly adopted metrics: Dice Loss [37], Intersection over Union (IoU) [38], Aggregated Jaccard Index (AJI) [39], and Panoptic Quality (PQ) [38]. Dice Loss and IoU measure semantic segmentation accuracy, while AJI and PQ are used to assess instance segmentation performance. Table 1 summarizes the test performance of our model on three benchmark datasets: NuInsSeg, MoNuSeg, and DSB2018. For the NuInsSeg dataset, we applied 5-fold cross-validation on 665 images collected from 31 organs. For the MoNuSeg dataset, the training set consists of 37 images from seven organs. The test set includes 14 high-resolution images ($3 \times 1000 \times 1000$ pixels), which were divided into 16 non-overlapping patches of size $3 \times 256 \times 256$ pixels to match the input size of the model. For the DSB2018 dataset, which includes 670 microscopy images, we split the data into 402 for training, 134 for validation, and 134 for testing.

Table 1 presents the quantitative performance evaluation of the proposed model. As illustrated in Table 1, the proposed model achieved strong and consistent results across all datasets and evaluation metrics. For semantic segmentation, the model achieved a Dice score of **0.8984** and IoU of **0.8232** on the NuInsSeg dataset, a Dice score of **0.9050** and IoU of **0.8317** on the MoNuSeg dataset, and a Dice score of **0.9355** and IoU of **0.8817** on the DSB2018 dataset. These high values reflect the model's ability to accurately align predicted masks with ground truth at the pixel level across diverse staining, tissue, and imaging conditions. For instance segmentation, the model achieved an average AJI of **0.7020** and PQ of **0.6807** on NuInsSeg, an AJI of **0.8078** with a PQ of **0.7414** on MoNuSeg, and further improved to an AJI of **0.8363** and PQ of **0.8071** on DSB2018. These results demonstrate the robust capacity of the model to effectively separate and delineate individual nuclei instances across diverse datasets, tissue types, and nuclei densities.

In terms of computational efficiency, the model maintains a low inference latency of approximately **0.0154 s per image**, uses **377.05 MB** of GPU memory, and contains **62.84 million parameters**. These statistics highlight the model's scalability and practicality for real-world, high-throughput applications without compromising accuracy.

4.5. Ablation

This section presents an ablation study on the NuInsSeg dataset [34], highlighting the effectiveness of our contributions in both the pre-processing and model design components to address critical challenges in histopathological image segmentation, particularly for densely packed nuclei. In the pre-processing pipeline, we proposed reference based style transformation for stain normalization and augmentation, detailed in Section 2.2, to standardize variability across images from different organs. Table 2 demonstrates that this technique significantly enhances segmentation performance by mitigating the impact of staining inconsistencies. On the model side, we introduce several key architectural components to improve feature extraction and segmentation accuracy. First, the integration of multi-dilated residual blocks into the U-Net architecture allows for the extraction of multi-scale features, which is critical for capturing intricate structures, such as densely packed and overlapping nuclei. Second, attention gates are incorporated into the skip connections of the U-Net. These gates refine the feature flow by focusing on relevant regions and suppressing irrelevant or noisy information, improving the model's ability to segment nuclei accurately. Finally, a bottleneck transformer is embedded in the bottleneck layer of the U-Net. This component enhances the network's ability to capture global contextual information, complementing the local feature extraction capabilities of the multi-dilated residual blocks. As evidenced in Table 2, these innovations collectively deliver significant improvements in segmentation performance. The integration of pre-processing and the proposed architectural components establishes a robust framework capable of achieving state-of-the-art results in challenging histopathological image segmentation tasks.

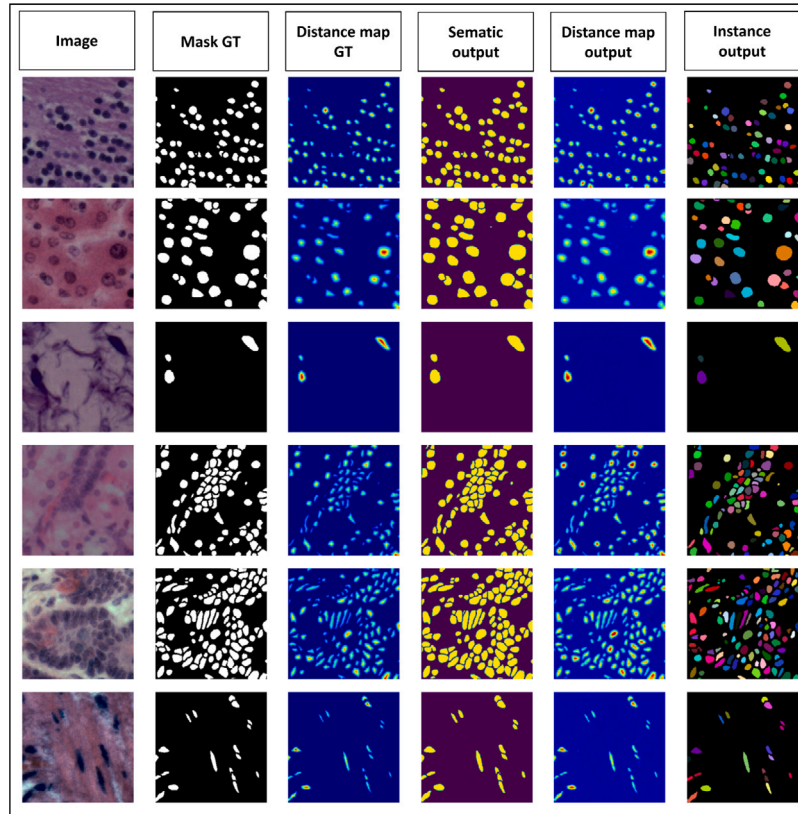


Fig. 7. Representative examples from the NuInsSeg dataset, showcasing a diverse selection of organ samples. From top to bottom, the figure displays segmentation results for the cerebellum, liver, kidney, bladder, jejunum, and cardia. These samples highlight the model's ability to effectively segment different tissue types across various organs.

Table 1
Performance and computational statistics across datasets and folds.

Dataset	Images & Organs	Fold	Dice	IoU	AJI	PQ	Time (s/img)	GPU (MB)	Params (M)
NuInsSeg	665, 31 organs	1	0.8926	0.8178	0.6890	0.6624	0.0153	394.52	62.84
		2	0.8996	0.8245	0.7126	0.6876	0.0151	394.77	62.84
		3	0.9012	0.8293	0.7266	0.7059	0.0159	394.52	62.84
		4	0.8946	0.8150	0.6827	0.6505	0.0157	394.77	62.84
		5	0.9039	0.8292	0.6990	0.6969	0.0152	394.52	62.84
		Avg	0.8984	0.8232	0.7020	0.6807	0.0154	394.62	62.84
			$\pm$ 0.0042	$\pm$ 0.0059	$\pm$ 0.0159	$\pm$ 0.0209	$\pm$ 0.0003	$\pm$ 0.11	
MoNuSeg	224, 7 organs	–	0.9050	0.8317	0.8078	0.7414	0.0166	394.52	62.84
DSB2018	134 images	–	0.9355	0.8817	0.8363	0.8071	0.0216	377.05	62.84

Table 2
Ablation study results on individual component impact: **Stage I:** Multi-dilated residual module, **Stage II:** Attention gate on the skip connection, **Stage III:** Reference based style transformation, **Stage IV:** Bottleneck transformer. $\uparrow$: larger is better.

Stage I	Stage II	Stage III	Stage IV	Dice loss $\uparrow$	IOU $\uparrow$	AJI $\uparrow$	PQ $\uparrow$
✓	x	x	x	0.6745	0.5257	0.6348	0.4885
✓	✓	x	x	0.7224	0.5767	0.6375	0.4861
✓	✓	✓	x	0.8311	0.7272	0.6721	0.6788
✓	✓	✓	✓	0.8984	0.8232	0.7020	0.6807

4.6. Comparison with existing methods

To thoroughly evaluate the effectiveness and generalizability of our proposed model, we conducted comparative experiments using three widely recognized datasets: NuInsSeg [34], MoNuSeg [35], and DSB2018 [36]. Each dataset was curated to address specific challenges

in histopathological nucleus segmentation and offers distinct advantages in terms of diversity, scale, and difficulty. The **MoNuSeg** dataset was developed as part of the 2018 Multi-Organ Nucleus Segmentation Challenge. Its goal was to encourage the development of generalizable models capable of performing nucleus segmentation across different tissue types. The **DSB2018** dataset was released for the 2018 Data Science Bowl hosted on Kaggle. It includes a collection of brightfield and fluorescence microscopy images from diverse biological sources, featuring nuclei of varying sizes, shapes, and levels of overlap. The dataset was designed to foster the development of segmentation algorithms that generalize well to unseen biological domains. Its wide modality range and variation in annotation quality make it particularly challenging. The **NuInsSeg** dataset was introduced in 2023 and published in Nature Scientific Data. It contains 665 image patches acquired at 40 $\times$ magnification from 21 human organs and 10 mouse organs, with more than 30,000 manually annotated nuclei. This diverse and consistently labeled dataset serves as one of the most comprehensive benchmarks for evaluating instance-level nucleus segmentation across

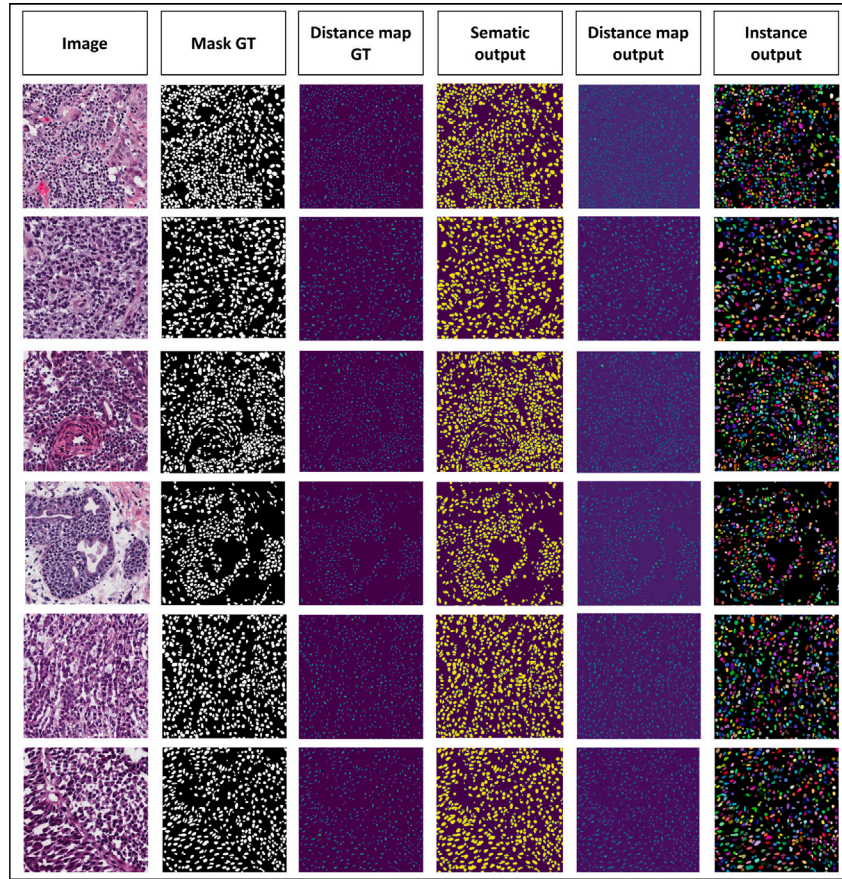


Fig. 8. Representative examples from the MoNuSeg test dataset, illustrating segmentation results for diverse organ types. From top to bottom, the figure includes samples from the breast, kidney, prostate, bladder, colon, and lung. These images demonstrate the model's robustness in segmenting various tissues across different organs, showcasing its generalizability and effectiveness in multi-organ segmentation tasks.

heterogeneous tissue types. Given the importance of fair benchmarking and meaningful evaluation, we structured our comparisons into two key parts: Comparison with Benchmark Methods and Comparison with SOTA Methods.

4.6.1. Comparison with benchmark methods

We first compared our model against baseline methods officially provided or recommended by the respective dataset organizers. For **NuInsSeg**, we compared our method with six commonly used benchmark models: Shallow U-Net [21], Deep U-Net [21], Attention U-Net [40], Residual Attention U-Net [41], Two-Stage U-Net [42], and Dual Decoder U-Net [43]. To ensure fairness, we followed the same training and validation protocols as used in the original studies [34]. Performance was evaluated using three standard metrics: Dice coefficient [37], Aggregated Jaccard Index (AJI) [39], and Panoptic Quality (PQ) [38]. As presented in Table 3, our model achieved the highest performance across all metrics, with a Dice score of 89.8%, AJI of 70.2%, and PQ of 68.1%. In contrast, the closest competitor, the Dual Decoder U-Net, reached 79.4% Dice, 55.9% AJI, and 51.3% PQ. This represents an improvement of over 10 percentage points in Dice, more than 14 percentage points in AJI, and nearly 17 percentage points in PQ. These gains demonstrate the superior segmentation accuracy and instance-level delineation achieved by our proposed architecture. We further evaluated our method using the **MoNuSeg** dataset and compared it to the top 10 ranked teams from the official challenge leaderboard. As reported in Table 4, our model achieved an AJI score of 0.8078, which outperforms all leaderboard submissions and reflects strong generalizability across heterogeneous tissue types and challenging conditions. In general, the high Dice score can be attributed to the

Table 3

Model performance comparison: The best values in the tables are highlighted in black bold black color. $\uparrow$: larger is better.

Model	Avg. Dice (%) $\uparrow$	Avg. AJI (%) $\uparrow$	Avg. PQ (%) $\uparrow$
Shallow U-Net [21]	78.8	50.5	42.7
Deep U-Net [21]	79.7	49.4	40.4
Attention U-Net [40]	80.5	45.7	36.4
Residual Attention U-Net [41]	81.4	46.2	36.9
Two-stage U-Net [42]	76.6	52.8	47.2
Dual Decoder U-Net [43]	79.4	55.9	51.3
Our model	89.8	70.2	68.1

architecture of our model, which incorporates multi-dilated residual blocks and attention mechanisms to improve the accuracy of semantics. Meanwhile, the improvements in AJI and PQ are primarily due to our post-processing strategy (Fig. 6). This strategy uses binary masks and distance maps to accurately separate overlapping nuclei. As a result, the model achieves more precise boundary delineation. This significantly enhances instance-level segmentation. The effect is especially evident in complex and densely packed tissue regions.

4.6.2. Comparison with SOTA methods

To benchmark our model against recent and advanced segmentation architectures, we selected a representative set of state-of-the-art (SOTA) methods with publicly available code and reproducibility support. These include DCAN [44], CRNCIS [45], ANCIS [46], Nucleiseg [47], Mask-RCNN [16], MS-RCNN [48], Cascade-MaskRCNN [49], HTC [50], Detectors [51], NucleiDetSeg [52], and MedSAM [53]. All models

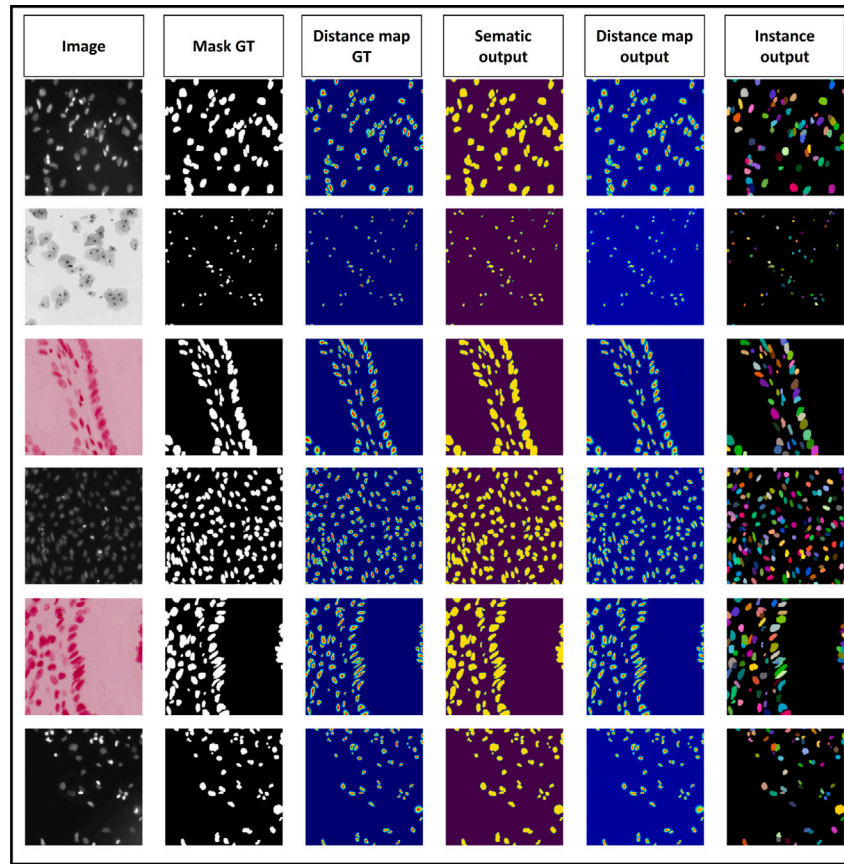


Fig. 9. Representative examples from the DSB2018 test dataset, illustrating segmentation results for diverse imaging types.

Table 4

MoNuSeg challenge team ranking and AJI performance.

Team	AJI $\uparrow$	Country	Rank
CUHK&IMSIGHT	0.6907	China	1
BUPT.J.LI	0.6868	China	2
Pku.hzq	0.6852	China	3
Yunzhi	0.6788	USA	4
Navid Alemi	0.6779	UK	5
Xuhuaren	0.6642	China	6
AetherAI	0.6632	Taiwan	7
Shuang Yang	0.6624	China	8
Bio-totem&SYSUCC	0.6619	China	9
Amirreza Mahbod	0.6574	Austria	10
Our team	0.8078	–	–

Table 5

Computational comparison on NVIDIA RTX A6000 GPU.

Model	Inference time (s)	GPU memory (MB)	Total parameters (M)
DCAN	0.009	220	16
CRNCIS	0.007	180	10
ANCIS	0.008	200	12
Nucleiseg	0.01	230	35
Mask-RCNN	0.018	1220	44
MS-RCNN	0.02	1230	45
Cascade Mask-RCNN	0.025	1400	55
HTC	0.03	1500	65
Detectors	0.028	1440	43
NucleiDetSeg	0.020	1240	60
MedSAM	0.775	2905	93
Our	0.015	377	62

were trained and evaluated under the same experimental settings. To benchmark our model against recent and advanced segmentation architectures, we selected a representative set of state-of-the-art (SOTA) methods with publicly available code and reproducibility support. These include DCAN [44], CRNCIS [45], ANCIS [46], Nucleiseg [47], Mask-RCNN [16], MS-RCNN [48], Cascade-MaskRCNN [49], HTC [50], Detectors [51], NucleiDetSeg [52], and MedSAM [53]. All models were trained and evaluated under the same experimental settings.

Quantitative analysis on the MoNuSeg dataset (Fig. 11) clearly shows that our approach surpasses existing methods across all metrics. Specifically, our method achieved scores of 63.0 AP@mask, 87.4 AP@0.5, 74.1 AP@0.75, and 80.8 AJI, clearly surpassing the closest competitor, NucleiDeSeg. These superior results highlight our model's enhanced capabilities in accurately identifying and segmenting nuclei. Visual analyses on the MoNuSeg dataset (Fig. 13) reinforce these quantitative findings. The qualitative comparison demonstrates our

method's ability to consistently provide precise delineation and effective separation of overlapping nuclei across diverse tissues, including breast, kidney, prostate, bladder, colon, lung, and brain. Notably, areas highlighted by the blue boxes indicate significant improvements in instance-level segmentation, where competing methods typically struggle. Similarly, quantitative comparisons on the DSB2018 dataset (Fig. 12) further confirm our method's superior performance, achieving 70.3 AP@mask, 91.2 AP@0.5, 81.9 AP@0.75, and 83.6 AJI. These scores consistently exceed those of all baseline methods. Qualitative visualizations in Fig. 14 illustrate our model's capability to accurately segment nuclei with improved boundary clarity and fewer segmentation errors, even in challenging imaging scenarios such as brightfield and fluorescence modalities.

Table 5 summarizes the inference time, GPU memory usage, and total parameters for various nucleus segmentation models evaluated on an NVIDIA RTX A6000 GPU for images sized 256×256 with batch

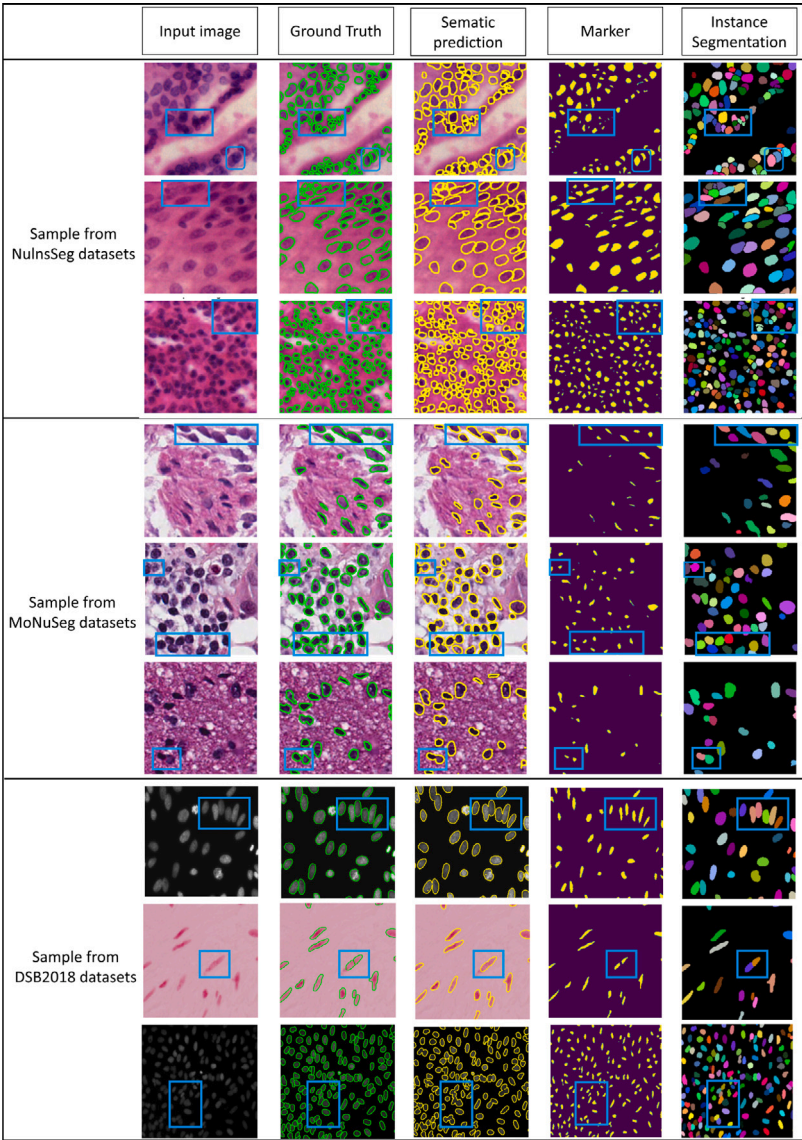


Fig. 10. Representative examples demonstrating the effectiveness of the proposed model in resolving overlapping nuclei across three benchmark datasets: NuInsSeg, MoNuSeg, and DSB2018. Each sample includes the input image, ground truth annotations, semantic segmentation output (highlighting the challenge of overlapping nuclei), marker maps generated from the distance map, and the final instance segmentation results. The blue boxes highlight regions with densely clustered or overlapping nuclei.

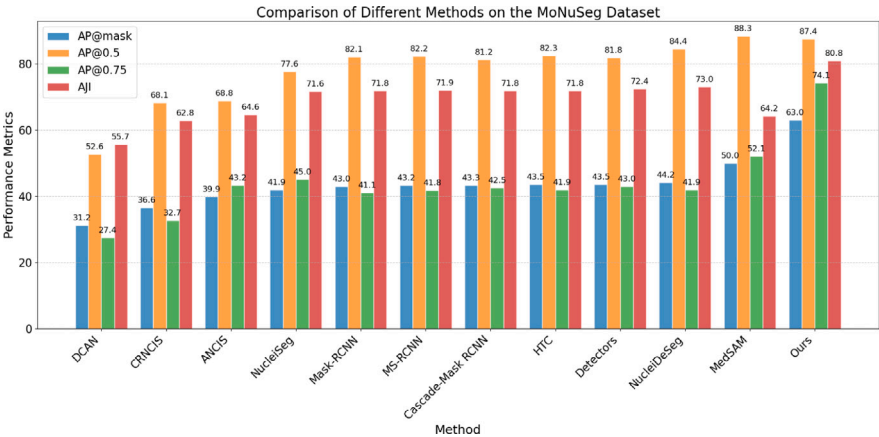


Fig. 11. Comparison of our method and the state-of-the-art methods on MoNuSeg.

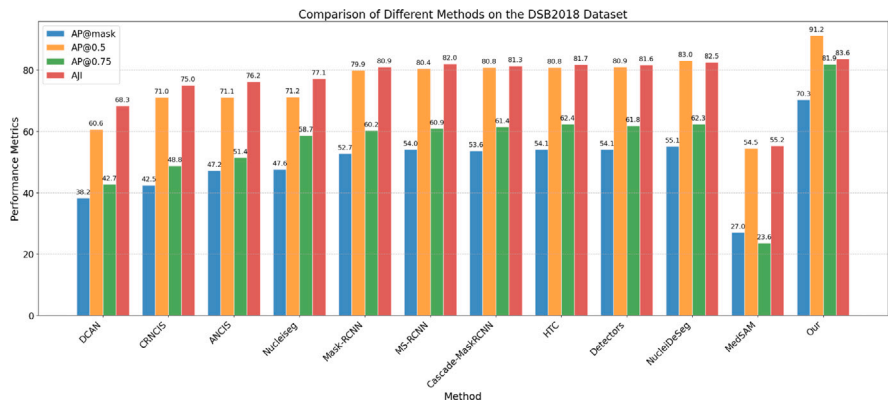


Fig. 12. Comparison of our method and the state-of-the-art methods on DSB2018.

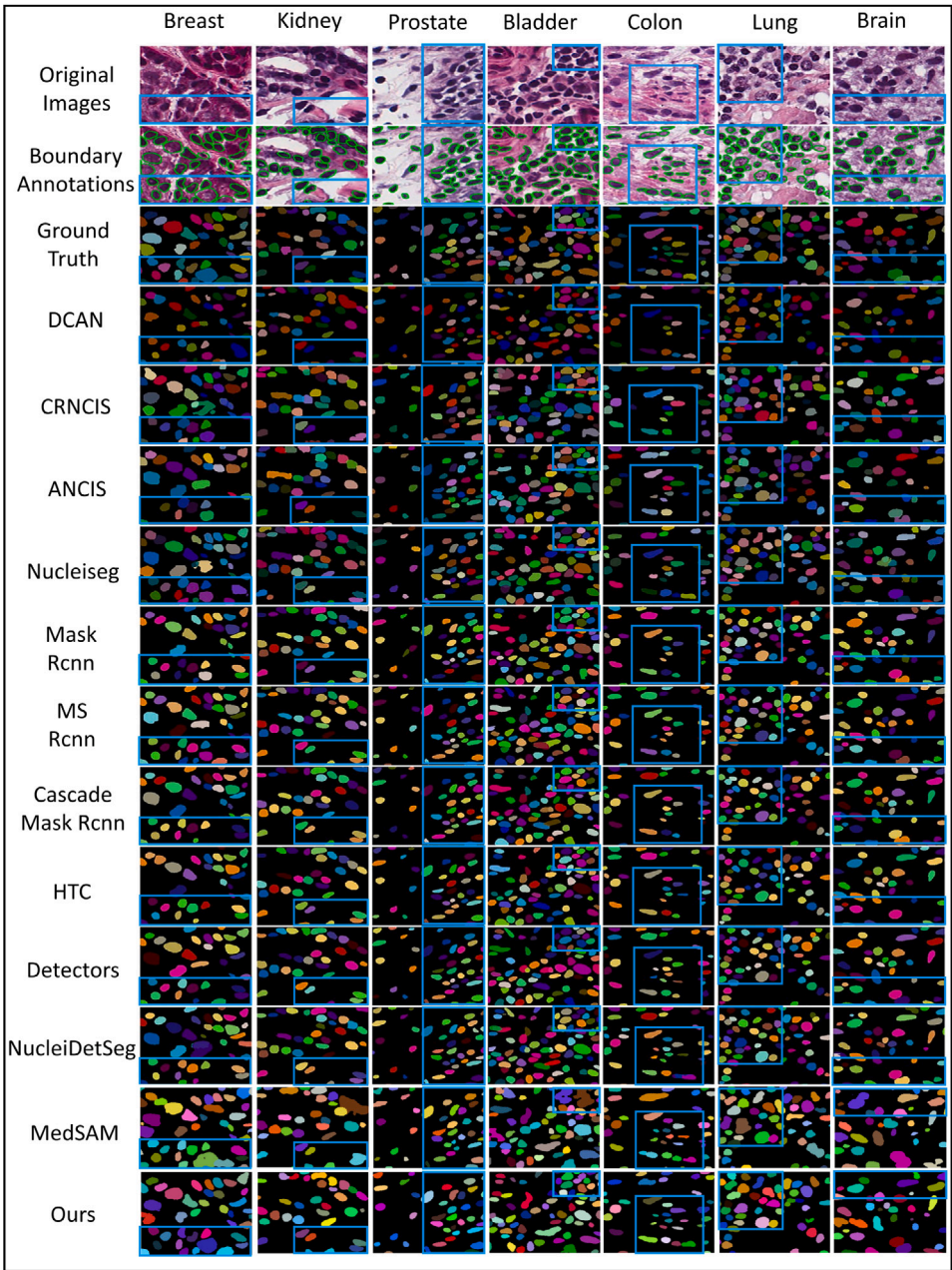


Fig. 13. Visual Comparison of our method and the state-of-the-art methods on MoNuSeg.

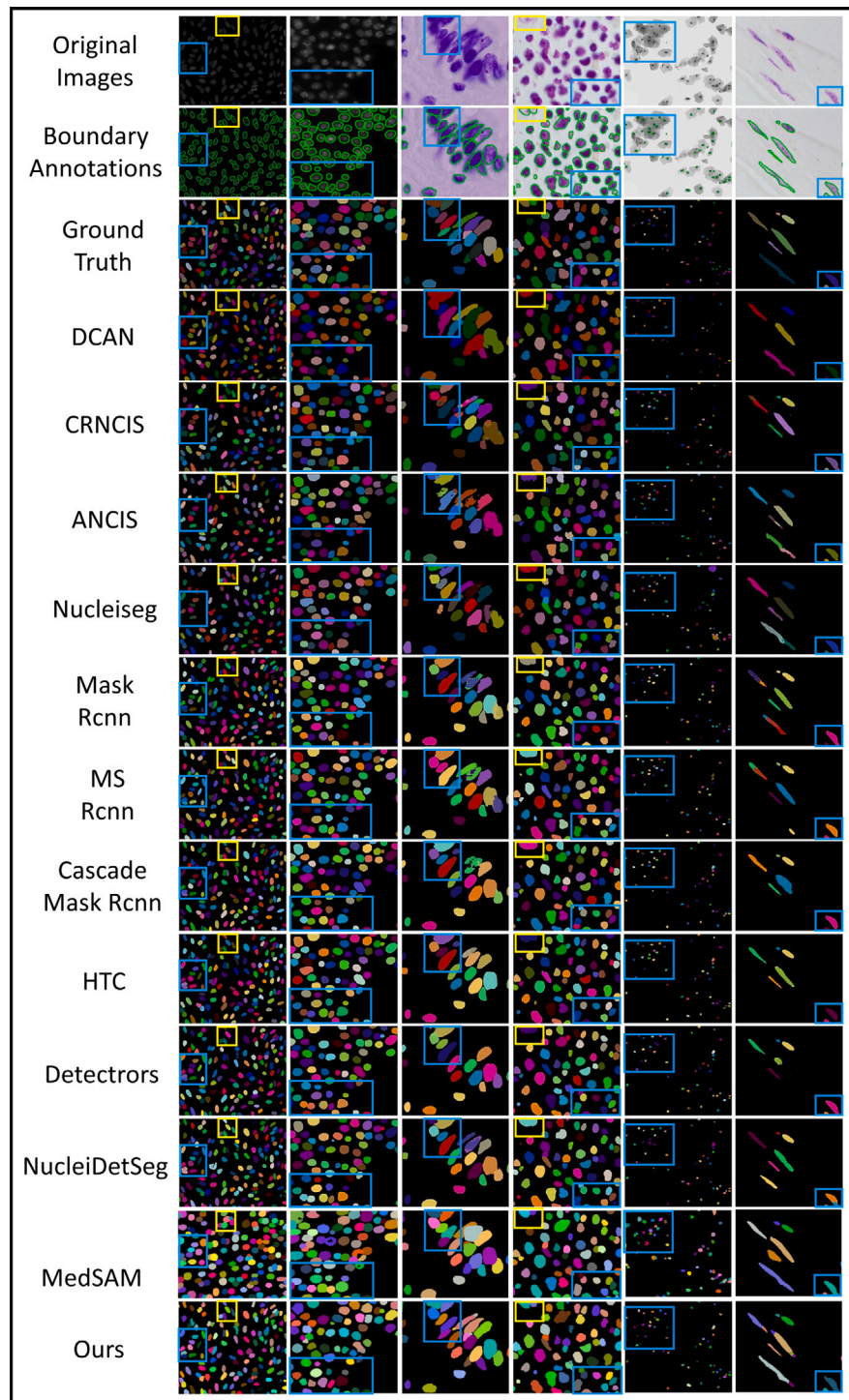


Fig. 14. Visual Comparison of our method and the state-of-the-art methods on DSB2018.

size of 1. Our proposed model (Attention-Based Multi-Dilated Residual U-Net with Transformer) achieves an inference speed (0.015 s/image) faster than several state-of-the-art detection-based methods such as Cascade Mask-RCNN (0.025 s), HTC (0.030 s), and Detectors (0.028 s). Although our model has a slightly higher GPU memory footprint (377 MB) due to advanced modules like transformers, it provides a competitive balance between speed and segmentation performance. Simpler CNN-only approaches like DCAN (0.009 s, 220 MB, 16 M params), CRNCIS (0.007 s, 180 MB, 10 M params) and ANCIS (0.008 s, 200 MB, 12 M params) offer lower latency and memory usage but often

struggle with densely overlapping nuclei. In the other hand, detection-based architectures such as Mask-RCNN (0.018 s, 1 220 MB, 44 M params), MS-RCNN (0.020 s, 1 230 MB, 45 M params) and NucleiDet-Seg (0.020 s, 1 240 MB, 60 M params) deliver similar speeds; however, they incur much larger memory footprints and are less effective at capturing global context, resulting in lower overall segmentation accuracy compared to our proposed model. Furthermore, MedSAM, a heavy and promptable ViT-based model with approximately 93 M parameters, achieves robust zero-shot segmentation but at the cost of 0.775 s per image and 2.9 GB GPU memory. Its high latency and memory demands

make it less practical for high-throughput histopathology workflows, although it excels at generalizing across unseen staining variations.

To conclude our contributions related to comparisons and achievements, we summarized the results in Figs. 7, 8, 9, 10, 11, 12, 13, and 14, as well as Tables 1, 3, 5, and 4. Figs. 7–10 and Table 1 illustrate that our model consistently demonstrated robust segmentation performance, showcasing its ability to generalize effectively across diverse organ types and staining variations. Additionally, Tables 3, and 4 and Figs. 11, 12, 13, 14 highlight that our method achieved state-of-the-art results on the NuInsSeg dataset, which includes data from 31 organs, and performs best on the MoNuSeg and DSB2018 datasets. Table 1 compares the time, GPU usage, and total parameters of existing methods with our proposed method. Overall, these findings highlight our model's adaptability and effectiveness with diverse multi-organ histopathological datasets. In the future, we plan to extend our method for nucleus and cell segmentation across multi-modal data, developing robust techniques to accurately segment nuclei and cells with varying shapes and appearances from different imaging modalities.

5. Conclusion

In conclusion, our proposed methodology successfully addresses the critical challenges of densely packed and overlapping nuclei segmentation in histopathological images. In this work we proposed multi-task learning framework incorporating it with advanced pre-processing techniques, such as reference-based style transfer with augmentation for domain generalization across diverse datasets. In the model architecture the multi-task encoder-decoder framework, equipped with attention gate-based multi-dilated residual blocks and a Bottleneck Transformer, effectively captures multi-scale features and preserves long-range dependencies. The integration of semantic segmentation and distance map generation, coupled with a watershed-based post-processing strategy, allows precise and robust instance segmentation. Our model demonstrated state-of-the-art performance on the NuInsSeg benchmark and, ranking 1st on the MoNuSeg benchmark with a 0.8078 AJI score. These outcomes highlight the adaptability and effectiveness of our approach for diverse multi-organ histopathological datasets with the satisfactory performance.

CRedit authorship contribution statement

Deboch Eyob Abera: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Nazar Zaki:** Writing – review & editing, Resources, Investigation. **Wenjian Qin:** Writing – review & editing, Writing – original draft, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared link to my data.

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